

Noah Sun

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Education

University of Waterloo – B.A.Sc. in Mechatronics Engineering (Candidate) September 2025 – April 2030
Relevant Coursework: Digital Computation, Algorithms and Data Structures, Circuits GPA: 3.92 / 4.00

Technical Skills

Languages: C/C++, Python, Java, JavaScript, TypeScript
Frameworks/AI: ROS2, OpenCV, TensorFlow, FastAPI, Express.js, React.js, Node.js, Flutter, Matplotlib
Data/Backend: PostgreSQL, MySQL, Microsoft Azure, Firebase, Pandas
Developer Tools: Git, Docker, Linux, GitHub Actions, Foxglove, REST APIs, Microsoft Power Apps, MS Office
Hardware: ESP32, Raspberry Pi 5, STM32 Nucleo, NVIDIA Jetson Nano, Arduino

Experience

Forward Deployed Engineer – Ascendance Foundry January 2026 – Current

- Built an end-to-end GTM tool combining web scraping with LLM-generated emails, increasing customers by **10%**
- Developed **3+ AI workflow automations**, introducing input validation and reducing human error by **15%**
- Designed ETL pipelines across Sage 50, VeriClock, and M365, delivering per-project statistics dashboards in Power BI

Embedded Flight Software Developer – Waterloo Aerial Robotics Group September 2025 – Current

- Led 4 engineers to develop a **32 MB flash driver** for config details and crash logs, streamlining debugging by **20%**
- Built an RTOS-integrated Flash Translation Layer with garbage collection and wear leveling, enhancing flash longevity

Co-founder and Lead Developer – Elapse | 🐙 [GitHub](#) June 2024 – August 2025

- Led 4 developers to build an iOS/Android app serving the 100k+ member VEX Robotics Competition community for real-time schedules, rankings, and match data
- Designed adaptive match times and scouting forms, contributing to **3500+ downloads** and **5.0-star rating**
- Integrated Firebase and Cloud Functions to sync data between team members, reducing scouting times by **30%**

V5RC Team Programming Lead – Checkmate Robotics | 📺 [Demo Video](#) | 🐙 [GitHub](#) June 2019 – August 2025

- Led team of 6 in robot design and library architecture, winning the **Create (innovation) Award** at VEX Worlds 2024
- Engineered ArkLib, a modular motion control library focused on scalability and flexibility using OOP, qualifying the team to VEX Worlds in **5 consecutive years** and winning the **Think (programming) Award** at international events
- Designed and tuned PID and control algorithms, achieving **1"/1° precision** movements across **300+ matches**

Vex AI Competition Team Embedded Systems Lead – Mi3L Schools May 2024 – June 2024

- Developed an API for the VEX Brain to process 3D spatial data from the Intel RealSense depth camera via the Jetson Nano for adaptive field navigation, winning the **Innovate Award** and **Skills World Champion** at VEX AI Worlds 2024

Projects

CityWatch | 🐙 [GitHub](#) March 2026

- Led a team of 4 to build a civic reporting platform analyzing **50k+ simulated reports** to identify high-risk zones
- Developed a **95%+ accurate** TensorFlow + OpenCV pipeline to classify road defects from citizen-submitted photos
- Generated **30-365 day infrastructure risk heatmaps** from spatial incident patterns and environmental features

VexDex | 🐙 [GitHub](#) August 2025 – Current

- Built full-stack competition analytics backend, managing **8000+ team records** and **40+ performance metrics**
- Engineered composite-key relational schema with targeted indexing, reducing database query time to **<100ms**
- Implemented dataset-level metadata logging to ensure atomic refresh tracking and API-visible data freshness

Autonomous Robot Navigation System | 📺 [Demo Video](#) | 🐙 [GitHub](#) September 2025

- Fused LiDAR and odometry data into a **0.1 m resolution costmap** for obstacle detection and robot localization
- Developed path planning and following using A*, Pure Pursuit, and PID controllers, achieving **0.1 m accuracy** and **flawless obstacle avoidance** across **50+ simulation tests**